

PARUL UNIVERSITY

Faculty of Engineering & Technology

B.Tech – Semester III

Subject: Design of Data Structures (303105201)

Time: 2.5 Hours

Maximum Marks: 60

SECTION – A

(Total: 30 Marks)

Q1 (Compulsory) – 12 Marks

a) Answer ANY THREE of the following: ($3 \times 2 = 6$)

1. Define Algorithm.
2. What is Stack Overflow?
3. Define Dynamic Memory Allocation.
4. What is a Deque?

b) Answer ANY THREE of the following: ($3 \times 2 = 6$)

1. Define Binary Tree.
 2. What is Hashing?
 3. Define Graph.
 4. What is Recursion Stack?
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Q2 – 9 Marks

a) Explain evaluation of postfix expression using stack. (4)

b) Answer ANY ONE of the following: (5)

- Evaluate the postfix expression:
6 2 3 + - 3 *
- Convert the infix expression $(A+B)*(C-D)$ into postfix form.

Q3 – 9 Marks

- a) Explain Merge Sort algorithm. (4)
- b) Answer ANY ONE of the following: (5)
- Sort the following array using Merge Sort and show all steps:
67, 25, 89, 12, 55
 - Explain time and space complexity of Merge Sort.

SECTION – B

(Total: 30 Marks)

Q4 (Compulsory) – 12 Marks

- a) Answer ANY THREE of the following: ($3 \times 2 = 6$)

1. Define Tree.
2. What is a Binary Search Tree?
3. Define Balance Factor.
4. What is BFS?

- b) Answer ANY THREE of the following: ($3 \times 2 = 6$)

1. Define DFS.
 2. What is Hash Table?
 3. Define Adjacency Matrix.
 4. What is Collision Resolution?
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Q5 – 9 Marks

- a) Explain deletion operation in Binary Search Tree. (4)
- b) Answer ANY ONE of the following: (5)

- Explain deletion cases in BST with example.
 - Explain inorder successor and inorder predecessor.
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Q6 – 9 Marks

- a) Explain Graph representations. (4)
- b) Answer ANY ONE of the following: (5)
- Explain BFS traversal with example.
 - Explain DFS traversal with recursive algorithm.